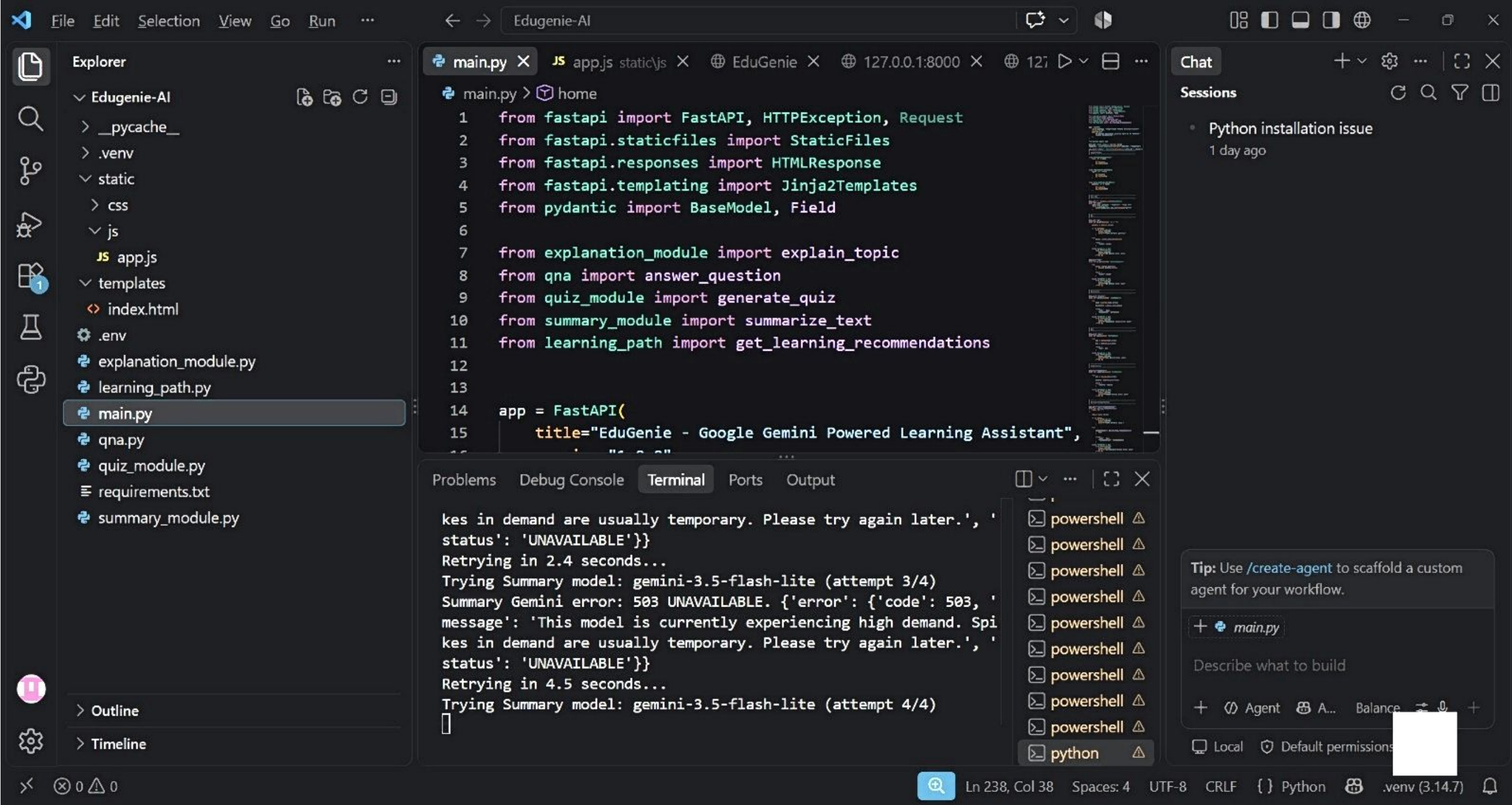




File

Edit

Selection

View

Go

Run

...

Edugenie-AI

...

qna.py

learning_path.py

quiz_module.py

summary_module.py

qna.py > answer_question

```
1 import os
2 import time
3 import random
4
5 from dotenv import load_dotenv
6 from google import genai
7
8 load_dotenv()
9
10 API_KEY = os.getenv("GEMINI_API_KEY")
11
12 if not API_KEY:
13     raise RuntimeError("GEMINI_API_KEY is missing from .env")
14
15 client = genai.Client(api_key=API_KEY)
```

Problems

Debug Console

Terminal

Ports

Output

kes in demand are usually temporary. Please try again later.', 'status': 'UNAVAILABLE'}}
Retrying in 2.4 seconds...
Trying Summary model: gemini-3.5-flash-lite (attempt 3/4)
Summary Gemini error: 503 UNAVAILABLE. {'error': {'code': 503, 'message': 'This model is currently experiencing high demand. Spi
kes in demand are usually temporary. Please try again later.', 'status': 'UNAVAILABLE'}}
Retrying in 4.5 seconds...
Trying Summary model: gemini-3.5-flash-lite (attempt 4/4)

powershell

powershell

powershell

powershell

powershell

powershell

powershell

python

Chat

Sessions

Python installation issue

1 day ago

Tip: Use /create-agent to scaffold a custom agent for your workflow.

+ qna.py

Describe what to build

+ Agent A... Balance

Local Default permissions

Edugenie-AI

static

css

js

app.js

templates

index.html

.env

explanation_module.py

learning_path.py

main.py

qna.py

quiz_module.py

requirements.txt

summary_module.py

Outline

Timeline

Ln 114, Col 6

Spaces: 4

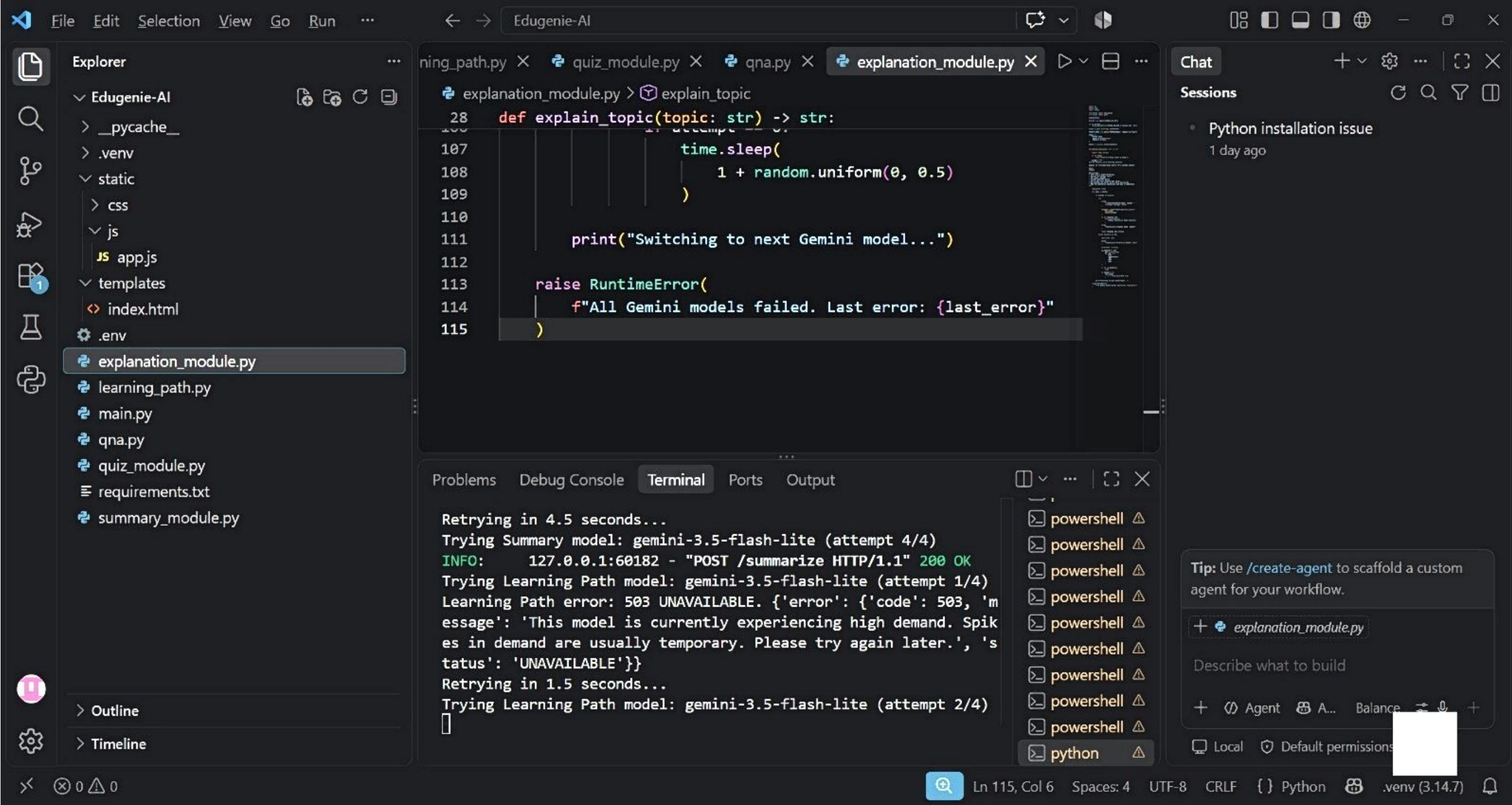
UTF-8

CRLF

{ }

Python

.venv (3.14.7)



File

Edit

Selection

View

Go

Run

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←

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Edugenie-AI

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Explorer

...

Edugenie-AI

> __pycache__

> .venv

> static

> css

> js

JS app.js

> templates

> index.html

.env

explanation_module.py

learning_path.py

main.py

qna.py

quiz_module.py

requirements.txt

summary_module.py

> Outline

> Timeline

ning_path.py

quiz_module.py

qna.py

explanation_module.py

...

Chat

More Actions...

Python installation issue

1 day ago

quiz_module.py > generate_quiz

1 import os

2 import json

3 import time

4 import random

5

6 from dotenv import load_dotenv

7 from google import genai

8

9 load_dotenv()

10

11 API_KEY = os.getenv("GEMINI_API_KEY")

12 MODEL = os.getenv("GEMINI_MODEL", "gemini-3.5-flash-lite")

13

14 if not API_KEY:

15 raise RuntimeError("GEMINI_API_KEY is missing from .env")

16

Problems

Debug Console

Terminal

Ports

Output

INFO: 127.0.0.1:60182 - "POST /summarize HTTP/1.1" 200 OK

Trying Learning Path model: gemini-3.5-flash-lite (attempt 1/4)

Learning Path error: 503 UNAVAILABLE. {'error': {'code': 503, 'message': 'This model is currently experiencing high demand. Spikes in demand are usually temporary. Please try again later.', 'status': 'UNAVAILABLE'}}

Retrying in 1.5 seconds...

Trying Learning Path model: gemini-3.5-flash-lite (attempt 2/4)

INFO: 127.0.0.1:57759 - "GET /learn/recommendations?topic=Artificial%20Intelligence HTTP/1.1" 200 OK

📄 powershell

📄 powershell

📄 powershell

📄 powershell

📄 powershell

📄 powershell

📄 powershell

📄 powershell

📄 python

Tip: Use /create-agent to scaffold a custom agent for your workflow.

+ quiz_module.py

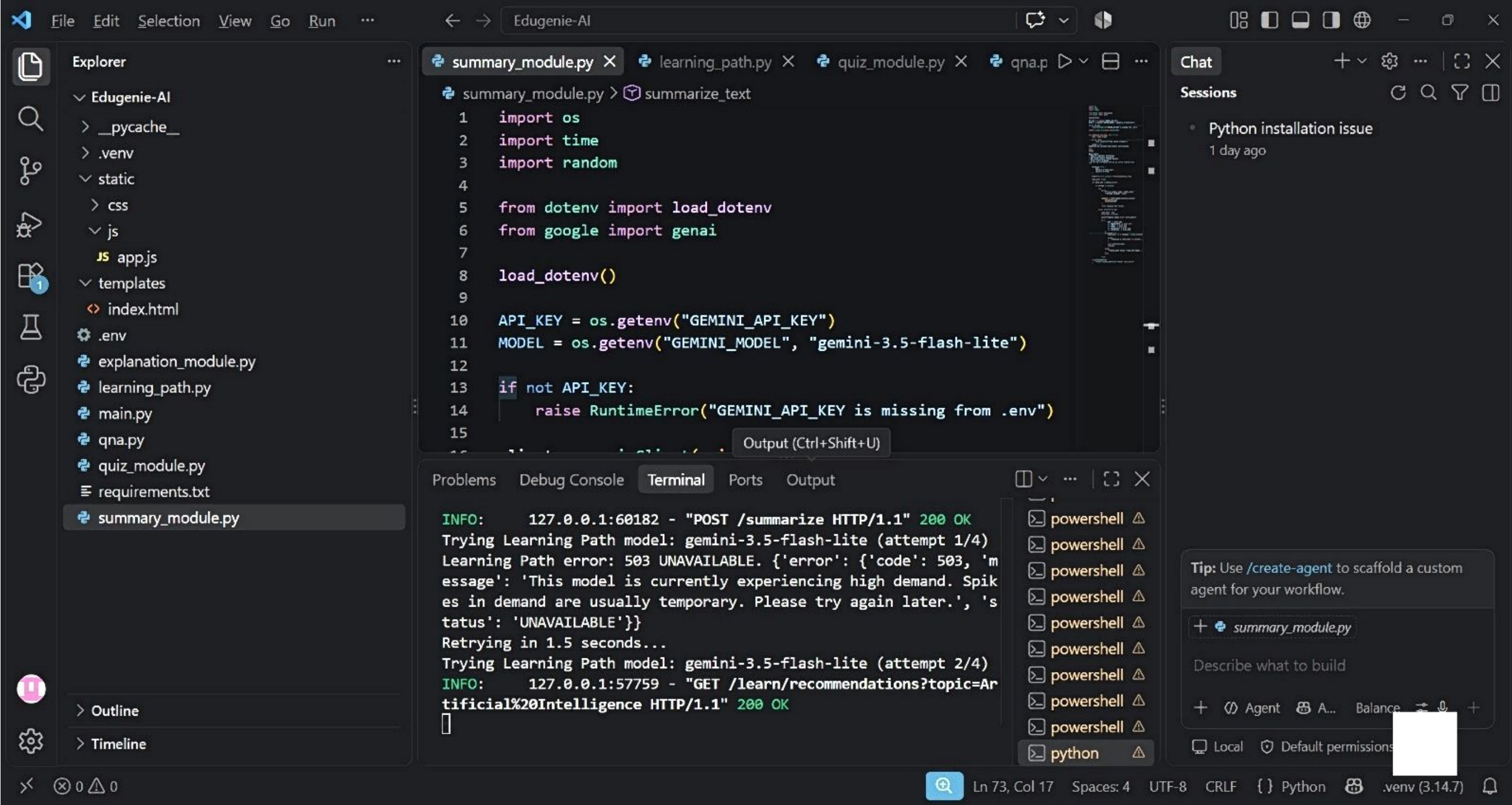
Describe what to build

+ Agent A... Balance

Local Default permissions

< 0 0

Ln 133, Col 6 Spaces: 4 UTF-8 CRLF {} Python .venv (3.14.7)



File

Edit

Selection

View

Go

Run

...

Edugenie-AI

127.0.0.1:8000

summary_module.py

learning_path.py

Chat

More Actions...

Python installation issue

1 day ago

Edugenie-AI

__pycache__

.venv

static

css

js

app.js

templates

index.html

.env

explanation_module.py

learning_path.py

main.py

qna.py

quiz_module.py

requirements.txt

summary_module.py

learning_path.py

get_learning_recommendations

1 import os

2 import time

3 import random

4

5 from dotenv import load_dotenv

6 from google import genai

7

8 load_dotenv()

9

10 API_KEY = os.getenv("GEMINI_API_KEY")

11 MODEL = os.getenv("GEMINI_MODEL", "gemini-3.5-flash-lite")

12

13 if not API_KEY:

14 raise RuntimeError("GEMINI_API_KEY is missing from .env")

15

Problems

Debug Console

Terminal

Ports

Output

INFO: 127.0.0.1:60182 - "POST /summarize HTTP/1.1" 200 OK

Trying Learning Path model: gemini-3.5-flash-lite (attempt 1/4)

Learning Path error: 503 UNAVAILABLE. {'error': {'code': 503, 'message': 'This model is currently experiencing high demand. Spikes in demand are usually temporary. Please try again later.', 'status': 'UNAVAILABLE'}}

Retrying in 1.5 seconds...

Trying Learning Path model: gemini-3.5-flash-lite (attempt 2/4)

INFO: 127.0.0.1:57759 - "GET /learn/recommendations?topic=Artificial%20Intelligence HTTP/1.1" 200 OK

powershell

powershell

powershell

powershell

powershell

powershell

powershell

python

Tip: Use /create-agent to scaffold a custom agent for your workflow.

+ learning_path.py

Describe what to build

+ Agent A... Balance

Local Default permissions

<

0

0

Ln 103, Col 6

Spaces: 4

UTF-8

CRLF

{ }

Python

.venv (3.14.7)

Ask a Question

What is Artificial Intelligence?

Ask Question

answer: Hello! I'm EduGenie, your learning assistant. I'd be happy to break this down for you. At a college level, **Artificial Intelligence (AI)** is best defined as a branch of computer science focused on building systems capable of performing tasks that typically require human intelligence. These tasks include learning, reasoning, problem-solving, perception, language understanding, and decision-making. Instead of just following rigid, pre-written instructions (like traditional software), modern AI systems are designed to process data, recognize patterns, and adapt based on new information. To understand AI fully, it helps to look at a few core concepts and how they fit together: ### 1. The Core Subsets of AI * **Machine Learning (ML):** This is the current engine driving most AI advancements. Instead of programming a computer with specific rules to solve a problem, you feed it massive amounts of data and let algorithms *learn* the rules on their own. * **Deep Learning:** A more advanced subset of machine learning inspired by the structure of the human brain (called artificial neural networks). It excels at processing complex, unstructured data like images, audio, and raw text. * **Natural Language Processing (NLP):** This allows computers to understand, interpret, and generate human language. It's the technology that powers translation tools and smart assistants. * **Computer Vision:** This gives machines the ability to "see" and interpret visual information from the world, such as digital images and videos. ### 2. Real-World Examples * **Recommendation Engines:** Platforms like Netflix, Spotify, or TikTok use ML algorithms to analyze your past behavior and predict what movies, songs, or videos you will want to see next. * **Generative AI:** Tools like ChatGPT or image generators (e.g., Midjourney) use deep learning to create entirely new content—essays, code, artwork, or music—based on prompts. * **Autonomous Vehicles:** Self-driving cars (like those from Waymo or Tesla) combine computer vision, sensor data, and real-time decision-making algorithms to navigate complex traffic environments without human intervention. In short, AI isn't just a single technology; it is a broad spectrum of techniques aimed at making computer systems smarter, more adaptable, and capable of augmenting human capability. Do you have any specific areas of AI you'd like to explore further, such as ethics, algorithms, or industry applications?



Explain a Topic

Artificial Intelligence

Explain

topic: Artificial Intelligence

explanation: Hello! I am **EduGenie**, your AI learning assistant. Today, we are going to break down one of the most transformative fields in modern technology: **Artificial Intelligence (AI)**. Don't worry if the term sounds intimidating—by the end of this guide, you'll have a clear understanding of what it is, how it works, and why it matters to you as a college student. --- **What is Artificial Intelligence?** At its simplest, **Artificial Intelligence (AI)** refers to the simulation of human intelligence in machines programmed to think, learn, and problem-solve like humans. Instead of just following strict, pre-written instructions (like a traditional computer program), AI systems can process vast amounts of data, recognize patterns, and make decisions or predictions on their own. --- **How Does AI Work? (The Core Concepts)** To understand AI, it helps to look at the three main pillars that power it: * **Data:** AI is data-hungry. Just as humans need to read books or practice skills to get smart, AI systems need massive amounts of data to learn. * **Algorithms:** These are sets of rules and mathematical formulas that tell the computer how to process the data and find patterns. * **Feedback Loops:** As the AI makes predictions or decisions, it checks its results. If it makes a mistake, it adjusts its internal settings to do better next time. This process is called **Machine Learning**. --- **Key Subfields of AI You Should Know** AI is a broad umbrella term. Underneath it, you will often hear about a few specific sub-branches: * **Machine Learning (ML):** A subset of AI where computers learn from data without being explicitly programmed for every single task. * **Deep Learning:** An advanced type of machine learning inspired by the human brain. It uses multi-layered "neural networks" to process complex data like images, audio, and natural language. * **Natural Language Processing (NLP):** The technology that allows computers to understand, interpret, and generate human language (this is how chatbots and translation tools work). --- **Real-World Example: Recommendation Algorithms** Think about the last time you used **Netflix, Spotify, or TikTok**. * **The Problem:** These platforms have millions of movies, songs, or videos. How do they help you find something you'll actually like? * **The AI Solution:** They use AI recommendation systems. Every time you watch a video, skip a song, or pause on a movie, the AI collects that data. It compares your behavior to millions of other users, recognizes patterns ("People who liked X and Y usually love Z"), and curates a personalized feed just for you. --- **Why AI Matters Today** * **Automation & Efficiency:** AI can handle repetitive, time-consuming tasks, freeing up humans to focus on creative and strategic work. * **Problem Solving:** AI helps scientists analyze complex medical scans for early disease detection, model climate change scenarios, and optimize global supply chains. * **Personalization:** From personalized learning apps to tailored shopping experiences, AI adapts to individual user needs. --- **EduGenie's Study Summary** * **Definition:** Machines designed to mimic human thinking and learning. * **Key Driver:** It relies on massive amounts of data and pattern recognition rather than rigid rules. * **Evolution:** Through Machine Learning and Deep Learning, AI systems improve automatically over time. * **Impact:** It shapes everything from entertainment algorithms to medical diagnostics. *** Do you have any questions about this topic, or would you like to explore a specific part of AI, like Machine Learning or Ethics in AI, next?

3. Which algorithm is commonly used for training neural networks by calculating the gradient of the loss function?

- ☐ Binary Search
- ☐ Backpropagation
- ☐ Merge Sort
- ☐ Dijkstra's Algorithm

Correct Answer: Backpropagation

4. What does NLP stand for in the context of Artificial Intelligence?

- ☐ Natural Language Processing
- ☐ Neural Logic Programming
- ☐ Network Layer Protocol
- ☐ Node Learning Process

Correct Answer: Natural Language Processing

5. Which type of machine learning involves an agent learning to make decisions by performing actions and receiving rewards or penalties?

- ☐ Supervised Learning
- ☐ Unsupervised Learning
- ☐ Reinforcement Learning
- ☐ Transfer Learning

Correct Answer: Reinforcement Learning

Artificial Intelligence Quiz

1. Which of the following is a subfield of machine learning that deals with algorithms inspired by the structure and function of the brain?

- ☐ Expert Systems
- ☐ Deep Learning
- ☐ Genetic Algorithms
- ☐ Robotics

Correct Answer: Deep Learning

2. What is the primary purpose of the Turing Test in artificial intelligence?

- ☐ To measure computational processing speed
- ☐ To evaluate a machine's ability to exhibit intelligent behavior indistinguishable from a human
- ☐ To test the energy efficiency of neural networks
- ☐ To determine the storage capacity of a database

Correct Answer: To evaluate a machine's ability to exhibit intelligent behavior indistinguishable from a human

3. Which algorithm is commonly used for training neural networks by calculating the gradient of the loss function?

Summarize

Artificial Intelligence (AI) is a technology that enables computers and machines to perform tasks that normally require human intelligence. These tasks include learning, problem-solving, decision-making, understanding language, and recognizing images. AI is widely used in applications such as virtual assistants, recommendation systems, healthcare, education, and self-driving vehicles. It helps automate tasks, improve efficiency, and provide intelligent solutions.

[Summarize](#)

Summary

Artificial Intelligence (AI) is a technology that lets computers and machines do tasks like humans, such as learning, solving problems, and recognizing images. It is used in areas like healthcare, education, virtual assistants, and self-driving cars to automate tasks, boost efficiency, and offer smart solutions.

Learning Recommendations

[Get Recommendations](#)



Learning Recommendations

Artificial Intelligence

Get Recommendations

topic: Artificial Intelligence

recommendation: Here is a clear and simple learning roadmap to help you master **Artificial Intelligence (AI)** from scratch! --- **### 1. Prerequisites (The Foundation)** Before jumping into AI, you need to brush up on a few basic subjects:

- Basic Python Programming:** Learn variables, loops, functions, and lists in Python (the main language for AI).
- High School Mathematics:**
 - Algebra:** Understanding functions and graphs.
 - Statistics & Probability:** Mean, median, standard deviation, and basic chances (crucial for how AI makes guesses).
 - Linear Algebra:** Basics of vectors and matrices (how computers store data).

--- **### 2. Beginner Topics (Getting Started)** Start with the basics of data and simple algorithms:

- Introduction to AI:** What is AI, Machine Learning (ML), and Deep Learning (DL)?
- Data Handling:** Learn how to use Python libraries like **Pandas** and **NumPy** to clean and organize data.
- Data Visualization:** Learn how to draw graphs using **Matplotlib** and **Seaborn** to see patterns in data.
- Supervised vs. Unsupervised Learning:** Learn the difference between teaching a computer with labeled answers vs. letting it find patterns on its own.

--- **### 3. Intermediate Topics (Core AI & Machine Learning)** Now you will start building actual intelligent models:

- Classic Machine Learning Algorithms:** Learn Linear Regression, Logistic Regression, Decision Trees, and K-Nearest Neighbors (KNN).
- Model Evaluation:** Learn how to check if your AI model is working well using terms like **Accuracy**, **Precision**, **Recall**, and **Train/Test Split**.
- Introduction to Deep Learning:** Learn how Artificial Neural Networks (ANNs) work, inspired by the human brain.
- Specialized AI Fields:**
 - Computer Vision (CV):** Teaching computers to "see" images and videos.
 - Natural Language Processing (NLP):** Teaching computers to understand human language (like text and speech).

--- **### 4. Advanced Topics (Cutting-Edge AI)** Take your skills to the pro level:

- Advanced Deep Learning Architectures:**
 - CNNs (Convolutional Neural Networks):** for advanced image recognition.
 - RNNs and Transformers (like GPT models):** for advanced text and language tasks.
- Generative AI (GenAI):** Learn how tools like ChatGPT, Midjourney, and Stable Diffusion work (Large Language Models and Diffusion Models).
- Reinforcement Learning:** Teaching an AI to make decisions by rewarding it for good actions (used in self-driving cars and game-playing bots like AlphaGo).
- AI Deployment:** Learn how to put your AI model on the web using **Flask**, **FastAPI**, or cloud services so other people can use it.

--- **### 5. Practical Projects (Build Your Portfolio)** Apply what you learn by building these fun projects:

- Beginner:**
 - House Price Predictor:** Predict house prices based on size and location using Linear Regression.
 - Spam Email Detector:** Classify emails as spam or not spam using text data.
- Intermediate:**
 - Cat vs. Dog Image Classifier:** Build a neural network that can tell if a picture is a cat or a dog.
 - Movie Review Sentiment Analyzer:** An app that reads a movie review and tells if the reviewer liked it or hated it.
- Advanced:**
 - Custom Chatbot:** Build a Question-Answering chatbot using a pre-trained Large Language Model API (like OpenAI).
 - AI Art Generator or Object Detector:** Build a system that detects objects in a live webcam feed.

--- **### 6. Practice Activities (Keep Your Skills Sharp)** Do these regularly to test yourself:

- Kaggle Competitions:** Join free data science competitions on Kaggle to solve real-world problems.
- Coding Challenges:** Solve Python problems on platforms like LeetCode and HackerRank.
- Read Research Papers:** Stay updated on the latest AI trends.